

USC ADVANCE Study — Threshold Paper: Analysis Results

Table 1a. Predictors of E-cigarette Use Subgroups — Users Only (USC ADVANCE Study)

Variable	No Exposure (n=510)	No Perceived Friends (n=226)	Neither (n=142)	Low Threshold (n=610)
Demographics				
Cohort C2	1.66	1.25	1.98*	1.11
Female	0.80	0.77	0.71	0.70
Hispanic/Latine	1.21	0.80	0.92	1.16
Asian	2.10	1.92	2.67*	2.16*
Sexual Minority	1.06	1.55	1.87*	1.27
Parent Education	0.98	0.90	0.88	0.97
GAD	1.12	1.24	1.16	0.90
MDD	1.07	0.57*	0.60	1.00
Grade Period Fixed Effects (ref = GP3)				
GP4	1.26	1.93	1.67	0.54
GP5	1.45	3.19*	3.35*	0.50
GP6	0.89	4.73**	3.39*	0.65
GP7	0.53	3.14	1.86	0.33
GP8	0.36	2.40	0.93	0.19*
Network Characteristics				
Out-degree (at)	0.54**	1.11	0.88	0.86
In-degree (at)	0.93	1.00	0.97	0.92
Perceived Friends' Use (TOA)	0.70**	—	—	0.76**
Network Exposure (TOA)	—	0.23**	—	—
Social Media				
sm_ecig_any (W4–W8)	1.38	0.52**	0.71	1.03

*Note. Values are ORs from logistic regression. *p < .05; **p < .01 (bold). All models include grade period fixed effects (factor(TOA), ref = GP3) and cohort dummy. No school fixed effects (separation problems in small subsets). Out-degree (at) and In-degree (at) are measured at TOA (time of adoption), consistent with the other TOA-anchored network terms. Subgroups are not restricted to non-isolates: students with zero ties (out-degree = 0) are retained throughout, since isolation is itself informative for several subgroups (e.g., isolates necessarily have zero network exposure and fall into Col 1). Only the threshold model in Table 2/3 excludes isolates, because threshold (k_{users}/n_{alters}) is undefined when $n_{alters} = 0$. Subgroup definitions at TOA — No Exposure: network exposure = 0; No Perceived Friends: perceived friend e-cig use = 0; Neither: both = 0; Low Threshold: $\leq 25\%$ of network friends using. Threshold cutoff: among the 841 adopters, 60.6% have a threshold of exactly 0, and the 70th percentile of the threshold distribution is 0.25; the 25% cutoff therefore captures adopters at or below the 70th percentile and is consistent with Valente (1996). — = excluded (variable directly defines subgroup, all values zero, or near-perfect separation). Perceived Friends' Use (TOA) excluded from Cols 2–3 (all values zero by definition). Network Exposure (TOA) excluded from Cols 1 and 3 (all values zero by definition) and from Col 4 because the low-threshold outcome is mechanically derived from the same exposure proportion. sm_ecig_any available W4–W8 only; GP3 adopters excluded from social media models.*

Table 1b. Predictors of Non-Use Subgroups — Resistant to Peer Influence (USC ADVANCE Study)

Variable	High Exposure (n=1,607)	High Perceived Friends (n=1,257)	Both High (n=793)
Demographics			
Cohort C2	0.79*	0.75**	0.87
Female	1.03	1.30**	1.63**
Hispanic/Latine	1.01	1.10	1.03
Asian	0.51**	0.34**	0.37**
Sexual Minority	0.78*	0.83	0.79

Parent Education	0.96	0.93**	0.98
GAD	0.92	1.08	0.93
MDD	1.20	1.51**	1.53**
Network Characteristics			
Out-degree (GP3)	1.39**	1.06*	1.25**
In-degree (GP3)	1.02	1.01	1.02
Friends (GP3)	1.13	—	2.03**
Friends (GP8)	1.47**	—	3.42**
Social Media			
sm_ecig_any (W4–W8)	—	—	—

*Note. Values are ORs from logistic regression. * $p < .05$; ** $p < .01$ (bold). Non-user subgroups — High Exposure: max network exposure > 0 ($n=1,607$); High Perceived Friends: max perceived friend use > 0 ($n=1,257$); Both High: both conditions met ($n=793$). Out-degree (GP3) and In-degree (GP3) are measured at baseline (study entry), since non-users have no time of adoption (TOA) to anchor to. Friends (GP3) is the baseline perceived friends' use measure; Friends (GP8) captures perceived friends' use at the end of the observation window and is the variable most directly tied to the High Perceived Friends / Both High subgroup definitions. As in Table 1a, subgroups are not restricted to non-isolates. Friends (GP3/GP8) excluded from Col5 (definitional overlap). sm_ecig_any not available for non-users (no time of adoption); shown as — for all columns.*

Table 2. Predictors of E-cigarette Adoption and Threshold — Main Model (GAD + MDD)

	Adoption N=3,530 (841 adopters, 2,689 non-adopters) 14,666 person-periods OR	Threshold (N=406) Coefficient
Constant	0.02**	-20.63
Grade Period (GP3 = reference)		
GP4	3.85**	17.23
GP5	24.38**	17.88
GP6	45.91**	18.11
GP7	74.56**	18.46
GP8	—	18.84
Cohort (C1 = reference)		
C2	1.19	0.97*
Attributes		
Female	1.12	0.26
Hispanic/Latine	0.85	0.07
Asian	0.68	-0.51
Parent Education	0.71	0.03
School Effects		
Student random intercept (1 id)	included	—
School random intercept (1 school)	included	—
School fixed effects (factor(school))	—	included
Mental Health		
GAD	0.31**	-0.03
MDD	3.44**	-0.21
Network Covariates		
Network exposure	6.46*	—
Friends who use e-cigarettes	1.64**	0.19**
Sexual minority	1.05	-0.12

Note. * $p < 0.05$; ** $p < 0.01$. Adoption: mixed-effects logistic regression with student (1|id) and school (1|school) random intercepts. $N = 3,530$ students (841 adopters, 2,689 non-adopters); 14,666 person-period observations at risk of adoption. School random intercept variance is estimated but individual school coefficients are not reported. GP8 near zero due to sparse events. Threshold: grouped binomial GLM with school fixed effects; excludes isolates ($n_alters = 0$); $N = 406$ non-isolate adopters. School 105 threshold omitted (near-perfect separation).

Table 3. Predictors of E-cigarette Adoption and Threshold — Sensitivity Model with School Fixed Effects (GAD + MDD)

	Adoption (Sensitivity) N=3,530 (841 adopters, 2,689 non-adopters) 14,666 person-periods OR	Threshold (N=406) Coefficient
Constant	0.02**	-20.63
Grade Period (GP3 = reference)		
GP4	1.00	17.23
GP5	1.02	17.88
GP6	0.67*	18.11
GP7	0.56**	18.46
GP8	—	18.84
Cohort (C1 = reference)		
C2	1.10	0.97*
Attributes		
Female	1.17	0.26
Hispanic/Latine	1.14	0.07
Asian	0.77	-0.51
Parent Education	0.95	0.03
School Fixed Effects (ref = School 101)		
School 102	0.60	0.16
School 103	0.76	0.03
School 104	1.17	1.09*
School 105	1.10	—
School 106	0.81	0.69
School 107	1.28	0.89
School 112	1.22	0.84
School 113	1.05	0.89
School 114	1.49	1.29**
Mental Health		
GAD	0.77*	-0.03
MDD	1.61**	-0.21
Network Covariates		
Network exposure	2.31**	—
Friends who use e-cigarettes	1.43**	0.19**
Sexual minority	1.18	-0.12

Note. * $p < 0.05$; ** $p < 0.01$. Adoption: standard logistic regression with school fixed effects (factor(school), ref = School 101) and grade period fixed effects; $N = 14,666$ person-period observations. Results are consistent with the main mixed model (Table 2) for key predictors. School 105 threshold coefficient omitted (near-perfect separation). This table serves as a sensitivity check allowing inspection of individual school effects.